

Modulation and Simulation

First Report – Modulation of a water distribution channel

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Abstract

This report presents an analysis of an open water distribution channel, focusing on the modulation of the flow of the water. In this work, it was simulated how the height of water in the channel changes when a gate in the end of the channel changes its level of aperture. These findings provide insight into the dynamic behaviour of open channels and serve as a foundation for future control and optimization studies.

Keywords

Equilibrium point, open channel, water flow, MATLAB simulation, Saint-Venant equations, Euler method.

Resumo

Este relatório consiste numa análise de um sistema aberto de distribuição de água, voltando o foco para a modulação de um fluxo de água. Neste trabalho, foi simulado como a altura da água no canal se altera com a mudança na abertura da comporta. As conclusões obtidas neste trabalho permitem compreender o comportamento dinâmico de um canal aberto, servindo de base para futuro controlo e otimização.

Palavras-chave

Ponto de equilíbrio, canal aberto, fluxo de água, simulação MATLAB, equações de Saint-Venant, método de Euler.

Table of Contents

Abstract	i
Resumo	ii
List of Figures	v
List of Tables	vi
List of Symbols	vii
1 Introduction	1
2 exercises 1, 2, 3 and 4	2
2.1 - Exercise 1	3
2.2 - Exercise 2	5
2.3 - Exercise 3	6
2.4 - Exercise 4	7
3 Conclusion	11
A1 Annex	15

List of Figures

Figure 2.2.1	5
Figure 2.3.1	6
Figure 2.4.1 a)	9
Figure 2.4.2 b)	10

List of Tables

Tabel 2.4.1 - Signs of the first and second derivative and concavity	10
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List of Symbols

h	Height of water in the channel in meters
t	Time in seconds
Q	Flow of water in liters per second
u	Gate aperture
g	Gravity acceleration
A	Area of the channel
b	Conversion unit's constant

List of Software

- MATLAB
(<https://www.mathworks.com/products/matlab.html>)

Chapter 1

Introduction

Water distribution channels are fundamental in our daily lives, for transporting water to one place where it is abundant to another where it is needed, whether it is in open channel, usually used in agriculture, or a closed channel, used in water distribution in urban areas. Modelling the channel must have this characteristic in account, since the mechanism that makes the water flow is not the same. For the first one is gravity, for the last one is pressure. With this thought in mind, to model the flow of water in an open channel, one might use the equations of Saint-Venant, which are partial derivatives, and have in account the principles of mass conservation and linear momentum. However, due to the complexity of the equations, it will be used simpler version, that is based in the assumption that the channel has a planar base, and the deep of the channel is always the same.

Chapter 2

Exercises 1, 2 ,3 and 4

In this chapter the exercises presented on the laboratorial guide will be solved and discussed.

2.1 - Exercise 1

Before proceeding with the exercise resolution, it is important to explain Euler's method and the rationale for its application. Euler's method is a simple and widely used numerical technique for solving ordinary differential equations. The fundamental concept is based on approximating the solution of an ordinary differential equation by advancing incrementally in time, under the assumption that the rate of change remains constant over small intervals Δt . Thus, utilizing differential equation, which models the water level and was provided in the laboratory guide [1], it is possible to rewrite it, allowing the obtain off the following equation.

$$\frac{dh(t)}{dt} = -\frac{ub\sqrt{2gh(t)}}{A} + \frac{10^{-3}Q}{A} \quad (2.1.1)$$

Where:

- Q is the flow of water [L/s]
- u is the aperture of the gate [%]
- g the acceleration of gravity, which is 9.8 [m/s²]
- $h(t)$ is the height of the water over time [m]
- A is the area of the channel [m²]
- b is a unit convert variable, equal to 0.00035 [m²]
- $\frac{dh(t)}{dt}$ as de derivative of the height as function of time

Knowing (2.1), by the definition of a derivative, the derivative of $h(t)$ can be expressed as:

$$\frac{dh(t)}{dt} = \lim_{\Delta t \rightarrow 0} \frac{h(t + \Delta t) - h(t)}{\Delta t} \quad (2.1.2)$$

We can solve the system represented in (2.1.1) by performing a discretization in time with an interval Δt . Substituting the continuous variable $h(t)$, with the discrete variable $x(k\Delta t)$, to represent samples of the signal $h(t)$ at periods defined by Δt , and if consider Δt sufficiently small to simulate a derivative, it is possible to take the following conclusions:

$$h(t) \approx x(k\Delta t) \quad (2.1.3)$$

$$\frac{dh(t)}{dt} = f(x(k\Delta t)) \quad (2.1.4)$$

For this reason, and using (2.1.2), we can write the following equation.

$$f(x(k\Delta t)) = \frac{x((k+1)\Delta t) - x(k\Delta t)}{\Delta t} \Leftrightarrow x((k+1)\Delta t) = \Delta t \cdot f(x(k\Delta t)) + x(k\Delta t) \quad (2.1.5)$$

We can, now, substitute (2.1.5) with the discretized form (2.1.1).

$$f(x(k\Delta t)) = -\frac{ub\sqrt{2gx(k\Delta t)}}{A} + \frac{10^{-3}Q}{A} \quad (2.1.6)$$

Substituting this expression into (2.1.5), it is obtained:

$$x((k+1)\Delta t) = \Delta t \cdot \left(-\frac{ub\sqrt{2gx(k\Delta t)}}{A} + \frac{10^{-3}Q}{A} \right) + x(k\Delta t) \quad (2.1.7)$$

This formula enables the representation, in an approximate manner, of the behaviour of the original continuous-time system by utilizing discrete time intervals Δt . From this result, it is concluded that predicting the subsequent state of the system from the current state is feasible, without requiring additional information about the past or the future.

2.2 - Exercise 2

Translating the equations from the previous question into a MATLAB program, it is obtained the code provided in [1]. In this code, after defining the variables (among them, Δt as 1 second, setting the time increments to 1 second steps, Q equal to 80, A equal to 8), there are two loops: the first one fills in the height values when the gate opening is at 50 percent, and the second one does so when the value increases to 55 percent. It is worth noting that, as required in the problem statement, a change in the gate opening only occurs once the response to the previous value has stabilized.

As a result of this experiment, there are obtained two plots: one representing the variation of the channel height over time and the other representing the variation of the gate opening with time.

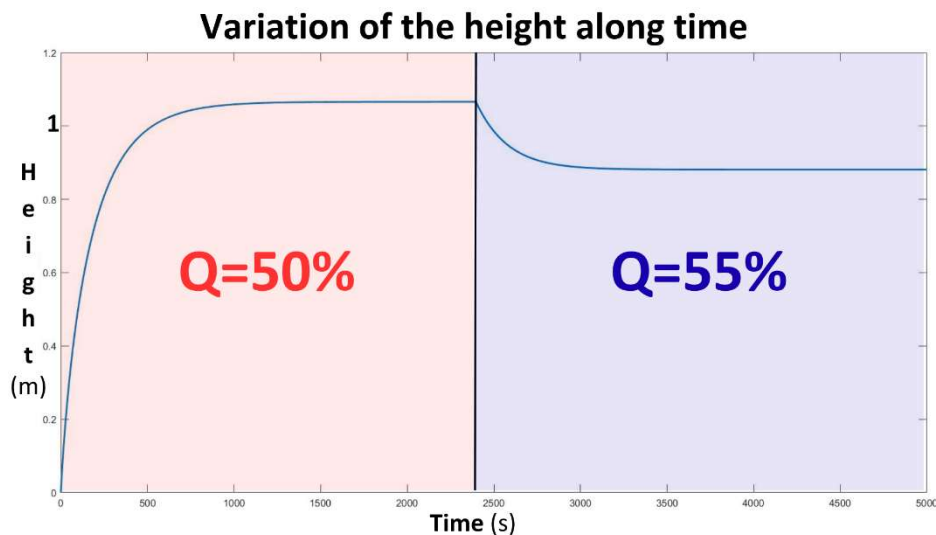


Figure 2.2.1 – Graphs obtained by simulation in MATLAB

As seen in Figure 2.2.1 the behaviour of this response, namely, its concavity and rate of change, will be analysed in more detail in later chapters, but it can be already noted that the responses are expected. Logically, when the gate opening increases, the height is expected to decrease (since more water leaves the channel, lowering its level). Another important remark is that the system is stable — something that can be observed in Figure 2.2.1. In both cases, the system's response tends to converge to an equilibrium point. When the system is subjected to a disturbance, it returns to its initial equilibrium point. On the other hand, if a change occurs in one of the system's parameters, it adjusts until it reaches a new equilibrium point.

2.3 - Exercise 3

In the graphs presented in figure 2.3.1, the variation of the water height (h) is observed for different initial conditions, considering that the gate opening remains constant over time, with $u = 50\%$. Values of height range from 0 (minimum water level) to 1,5 meters (maximum allowed level, as described in the problem statement). To analyse the system's behaviour, it is introduced the concept of the equilibrium point, which corresponds to the value of height at which the derivative is zero. At this point, the height stabilizes, as there is no longer any variation over time. By analysing the graphs obtained in MATLAB, it is observed that the system reacts differently under various initial conditions. However, regardless of the initial condition, the system always converges to the same point: the equilibrium point. The time required to reach this equilibrium varies — it may be faster or slower depending on the initial condition. Based on this graphical analysis, the following conclusions can be drawn:

- For initial conditions below the equilibrium point ($h < 1,0662$), the initial water height is lower than the equilibrium point. A progressive increase in height is observed over time until the system reaches a stable equilibrium state.
- If the initial conditions are above the equilibrium point ($h > 1,0662$), the initial water height is higher than the equilibrium point. The system responds with a gradual decrease in height, later stabilizing at a lower value corresponding to the equilibrium point.
- If the initial condition is at the equilibrium point ($h = 1,0662$), the height remains constant over time, since the derivative is zero from the beginning and there is no variation in height. This topic will be explained in more detail in next question.

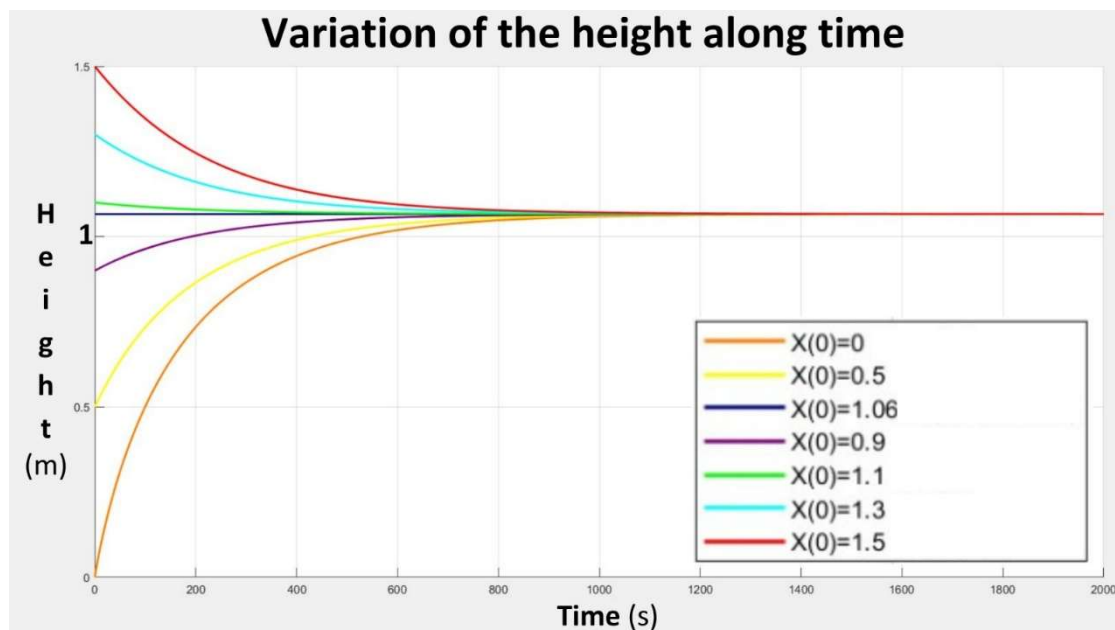


Figure 2.3.1 - Different responses for different initial conditions

2.4 - Exercise 4

Before discussing some relevant aspects concerning the differential equation that models the water level, it is necessary to draw some conclusions regarding the variables that compose it. In this sense, it is useful to first understand that one is dealing with a system whose response is essentially described by an initial condition, which may be above or below the equilibrium point, towards which the system converges over time (as seen in previous simulations), being reached as $t \rightarrow \infty$. This equilibrium level can be obtained by determining the water level for which its variation over time becomes zero, i.e., when $\frac{dh}{dt} = 0$, yielding the following expression:

$$-u \frac{b\sqrt{2g}}{A} \sqrt{h_{eq}} + \frac{10^{-3}}{A} Q = 0 \Leftrightarrow h_{eq} = \left(\frac{10^{-3}Q}{ub\sqrt{2g}} \right)^2 \quad (2.4.1)$$

Where:

- h_{eq} is the height of the water when the equilibrium is reached [m]

To make the understating of (2.4.1) easier, it is defined that:

- First term of (2.4.1) is $-u \frac{b\sqrt{2g}}{A} \sqrt{h_{eq}}$
- Second term of (2.4.1) is $\frac{10^{-3}}{A} Q$

Thus, it is readily observed that increasing the gate opening, u , results in a lower stabilized water level, while the opposite occurs for the flow rate, Q , where increasing its numerical value produces a higher stabilized water level, as observed in Figure 2.4.1. It is also important to see that the cross-sectional area, A , does not influence the equilibrium point at all. However, all these quantities influence the derivative of the height ($\frac{dh}{dt}$), i.e., the variation of height over time, altering the rate at which a point close to equilibrium is reached. In this sense, increasing the magnitude of the derivative will result in a more pronounced variation in the system's response, and decreasing it will result in a more gradual variation in the system's response.

The growth or decay of the system's response function, until a point close to equilibrium is reached, is logically determined by the sign of the derivative (positive: growth / negative: decay). With these aspects in mind, for the values chosen in the previous items, namely gate opening $u = 50\%$ and flow rate $Q = 80 \text{ L/s}$, it is known that, regarding the early mentioned derivative, the positive first term of (2.4.1) is greater in magnitude than the magnitude of the negative second term of (2.4.1) generating an increasing response function (as seen in previous questions). In this context, h_{eq} is the height at the equilibrium point, in meters.

From the analysis of figure 2.4.2, one can easily understand that increasing the value of the cross-sectional area A attenuates the magnitude of both mentioned terms of the derivative, reducing their numerical value and resulting in a smoother curve. Conversely, reducing the value of the cross-sectional area A amplifies the magnitude of the components, and consequently the value of the derivative, generating a steeper curve, reaching a point close to stabilization earlier.

From a dynamic perspective, it can also be interesting to understand the response graph obtained in figure 2.1, in the sense that starting from an initial situation of zero height, it is understood that, at an initial moment, the second term of (2.4.1) assumes a reduced value, so the growth rate of the water level is dependent on the flow rate, Q , and the cross-sectional area, A , this being the phase of greatest growth. As the height increases, the second term of (2.4.1) increases in magnitude, attenuating the growth component (first term of (2.4.1)), resulting in progressively slower growth.

Finally, one can also evaluate the differential equation that models the water level with respect to its second derivative, and the implications it brings to the behaviour of the response function of the system under study. Thus, it is obtained the following expression for the second derivative using the chain rule:

$$\frac{dh^2}{dt^2} = -\frac{1}{2}u \frac{b\sqrt{2g}}{A\sqrt{h_{eq}}} \frac{dh}{dt} \quad (2.4.2)$$

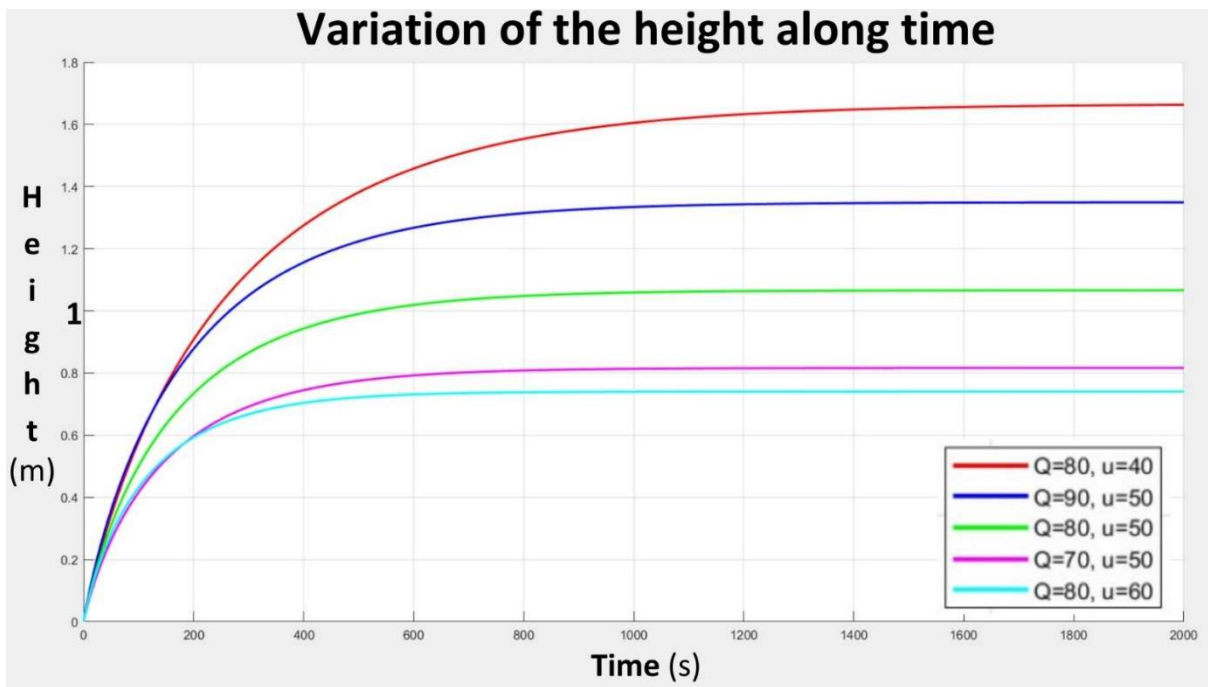
From this expression, given that all the parameters composing it assume a positive numerical value, one can conclude that, for any possible value of the water height, the second derivative will assume a value whose sign is determined by the inverse of the sign of the derivative, which has been studied previously. Thus, it becomes clear that for cases where the initial condition is less than the equilibrium state value, the response function will be increasing, and therefore derivative will be positive and the second derivative negative, generating a response function with downward concavity. For cases where the initial condition is greater than the equilibrium state value, the response function will be decreasing and therefore the derivative will be negative and the second derivative positive, generating a response function with upward concavity.

All these results are clear in Figure 2.3.1, knowing that in that exercise the equilibrium point, P_e , is 1,0662 meters.

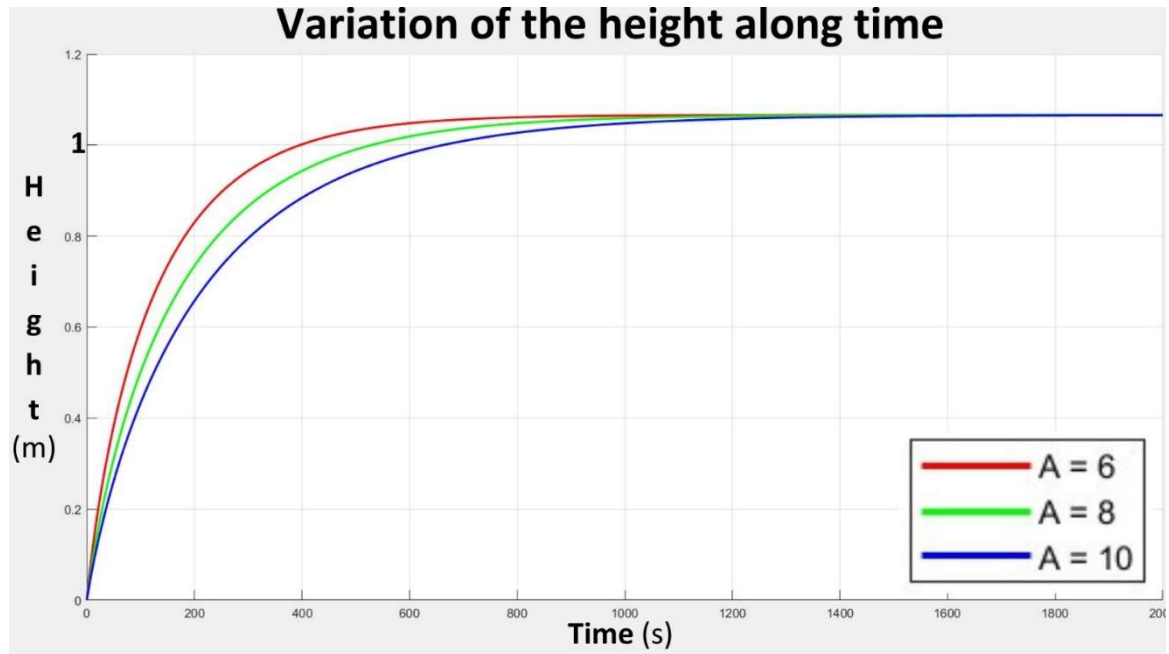
Table 2.4.1: Signs of the first and second derivative and concavity shape.

Variable	Signal				
$h(t)$	0	...	$p_e = \left(\frac{10^{-3}Q}{ub\sqrt{2g}}\right)^2$	...	h_{max}
$\frac{dh}{dt}$	+	+	0	−	−
$\frac{d^2h}{dt^2}$	−	−	0	+	+
Shape	∩	∩	Inflection Point	∪	∪

It is considered more appropriate to evaluate only the values in the interval $[0; h_{max}]$ ($h_{max} = 1.5m$), where h_{max} is the maximum high permitted, which is 1.5 meters since the water channel, represented by the function $h(t)$, is limited to these two values.



(a) Influence of area on the height of the water



b) Influence of water flow and gate opening on the height of the water

Figure 2.4.1 – Influence of the modelling parameters on the height of the water

Chapter 3

Conclusions

This report presented the modelling and simulation of an open water distribution channel, aiming to understand how the water level evolves over time under different operating conditions. Using a simplified differential equation derived from the Saint-Venant model and applying Euler's numerical method, the system's behaviour was successfully simulated in MATLAB. The results demonstrated that the channel behaves as a stable dynamic system, always converging to an equilibrium point that depends on the gate aperture and the water flow. Increasing the gate opening leads to a lower steady-state water level and a faster system response, while increasing the inflow raises the equilibrium height. The cross-sectional area affects the rate of variation but not the equilibrium value itself.

From the analysis of the first and second derivatives, it was also concluded that the magnitude of these derivatives increases with higher gate openings, leading to steeper and quicker responses toward equilibrium. This study provided a clear understanding of how the channel's parameters influence its dynamic behaviour and established a solid basis for future work focused on implementing automatic control strategies or extending the model using the full Saint-Venant equations.

Annexe

Theory about the model of an open channel and simulation

In this annexe, will be presented some of the equations behind the model of an open channel, and the MATLAB code.

A.1 Theory

As mentioned before in the introduction, the equations that rule the system of an open channel are the Saint-Venant equations, that will not be used due to their complexity. Instead, it will be used the equation A.1, that as the assumptions described earlier.

$$\frac{dh}{dt} = -u \frac{b\sqrt{2g}}{A} \sqrt{h} + \frac{10^{-3}}{A} Q \quad (\text{A.1})$$

A.2 MATLAB

```
%code related to laboratory 1 of the Msim course

%definition of variables

A = 8;           %area is 8 m^2
Q = 80;          %flow rate is 80 L/m
h_max = 1.5;     %maximum allowed height
g = 9.8;         %gravity acceleration (in m/s^2)
b = 0.00035;     %unit conversion constant in m^-2
u = 50;          %gate opening (50%) (note: keeping flow at 80, 42% is the
maximum)
interval = 1;    %time interval

%apply Euler method
tempo = 0:interval:5000; %define time (starts at 0 and goes up to 5000 in
intervals of 1 second)
h = zeros(size(tempo)); %output vector has the same size as time
initial_value = 0;      %we assume the initial value is 0 (as stated in the
problem)
h(1) = initial_value;   %so the initial height will also be 0

equilibrium_point = ((Q*0.001)/(u*b*sqrt(2*g)))^2; %equilibrium point of the
system
equilibrium_point_50 = equilibrium_point;           %relative to u=50
vector_u = 0:5000; %define vector for the
gate
```

```

for i = 1:(length(tempo) - 1)
    h(i+1) = h(i) + (interval * altura(h(i), A, b, u, g, Q)); %apply Euler
method
    vector_u(i) = u; %fill the gate
opening vector
    if abs(h(i+1) - equilibrium_point) < 1e-5 %if the difference between the
height value is less than 0.00001
%we consider that equilibrium is
reached (since
%it is reached at infinity)
        new = tempo(i); %instant when it stabilizes
        break %end the loop
    end
end

u = 55; %change the gate opening
equilibrium_point = ((Q*0.001)/(u*b*sqrt(2*g)))^2; %equilibrium point of the
system

for i = new:(length(tempo) - 1) %same loop as above, but here I go until the
end of time
    h(i+1) = h(i) + (interval * altura(h(i), A, b, u, g, Q));
    vector_u(i+1) = u;
end

%plot routines
gg = plot(tempo, h);
set(gg, 'LineWidth', 1.5);
title('Height variation over time'); %Graph title
xlabel('time [s]'); % X-axis label
ylabel('height [m]'); % Y-axis label

figure();
bg = plot(tempo, vector_u);
set(bg, 'LineWidth', 1.5);
ylim([40 60]); %Limit Y-axis between 0 and 100
title('Gate opening variation over time'); %Graph title
xlabel('time [s]'); % X-axis label
ylabel('percentage of gate open [%]'); % Y-axis label

%define the function
function dh = altura( h, A, b, u, g, Q)
    % Calculates the height variation (dh/dt)
    dh = (((-u*b*sqrt(2*g) * sqrt(h)) / A) + (0.001 * Q / A));
end

```

References

- [1] *Modelação de um canal de distribuição de água*, João Miranda Lemos, Modulação e simulação - Mestrado Integrado em Engenharia Eletrotécnica e de Computadores 2024/2025, Av. Rovisco Pais 1, 1049-001, Lisboa
- (<https://fenix.tecnico.ulisboa.pt/downloadFile/3096843219129383/MSim-2024-2025-Lab1.pdf>)